ME 333

MECHANICS OF MATERIALS

Lecture 8: Mohr's Circle for 3D stress state & Equilibrium Equations

Ravi Sastri Ayyagari

Mechanical Engineering
Indian Institute of Technology Gandhinagar

Sem I, 2025-2026

Mean and Deviatoric Stresses

- Mean stress can be defined as:

$$\sigma_m = \frac{1}{3}I_1 = \frac{\sigma_{xx}+\sigma_{yy}+\sigma_{zz}}{3} = \frac{\sigma_1+\sigma_2+\sigma_3}{3}$$

- Stress tensor can be additively decomposed into mean and deviatoric components.

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_m + \boldsymbol{\sigma}_d$$

- Deviatoric and mean stress tensors can then be defined as:

$$\boldsymbol{\sigma}_d = \begin{bmatrix} \sigma_{xx}-\sigma_m & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy}-\sigma_m & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz}-\sigma_m \end{bmatrix}; \quad \boldsymbol{\sigma}_m = \begin{bmatrix} \sigma_m & 0 & 0 \\ 0 & \sigma_m & 0 \\ 0 & 0 & \sigma_m \end{bmatrix}$$

Characteristics of Deviatoric Stress Tensor

- $\boldsymbol{\sigma}_d$ is defined as:

$$\boldsymbol{\sigma}_d = \begin{bmatrix} \sigma_{xx}-\sigma_m & \sigma_{xy} & \sigma_{xz} \\ \sigma_{xy} & \sigma_{yy}-\sigma_m & \sigma_{yz} \\ \sigma_{xz} & \sigma_{yz} & \sigma_{zz}-\sigma_m \end{bmatrix}$$

- $\boldsymbol{\sigma}_d$ represents a state of pure shear. The principal values of deviatoric stress tensor are:

$$S_1 = \sigma_1 - \sigma_m = \frac{2\sigma_1-\sigma_2-\sigma_3}{3} = \frac{(\sigma_1-\sigma_3)+(\sigma_1-\sigma_2)}{3}$$

$$S_2 = \sigma_2 - \sigma_m = \frac{(\sigma_2-\sigma_3)+(\sigma_2-\sigma_1)}{3}$$

$$S_3 = \sigma_3 - \sigma_m = \frac{(\sigma_3-\sigma_1)+(\sigma_3-\sigma_2)}{3}$$

- The trace of $\boldsymbol{\sigma}_d$ is zero. Therefore, only two Principal stress values are independent.
- The Principal directions of $\boldsymbol{\sigma}_d$ are same as that of $\boldsymbol{\sigma}$

Invariants of Deviatoric Stress Tensor

- Invariants of $\boldsymbol{\sigma}_d$ are:

$$J_1 = 0$$

$$J_2 = -\frac{1}{6}\left[(\sigma_{xx}-\sigma_{yy})^2+(\sigma_{yy}-\sigma_{zz})^2+(\sigma_{zz}-\sigma_{xx})^2+6(\sigma_{xy}^2+\sigma_{yz}^2+\sigma_{zx}^2)\right]$$

$$= -\frac{1}{6}\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right] = I_2 - \frac{1}{3}I_1^2$$

$$J_3 = (\sigma_1-\sigma_m)(\sigma_2-\sigma_m)(\sigma_3-\sigma_m) = \det(\boldsymbol{\sigma}_d)$$

- Von-Mises stress is defined as:

$$\sigma_{VM} = \sqrt{3|J_2|}$$

Octahedral Plane

- Plane which is oriented making equal angles with the principal axes (X, Y, Z) is called the *Octahedral Plane*.
- Normal to Octahedral plane is $N_{Oct} = \left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$
- Normal and shear stresses on Octahedral plane:

$$\begin{aligned}\sigma_{Oct} &= \frac{1}{3}I_1 = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)\\ \tau_{Oct} &= \sqrt{\frac{2}{3}|J_2|}\\ &= \frac{1}{3}\sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}\end{aligned}$$

Normal and Shear stresses in octahedral plane are also invariant

Plane Stress

- Consider the deformations occuring in thin flat plates. If b is the smallest inplane dimension, then:
  - Thin plates : $t < (b/20)$
  - Thick plates: $t > (b/20)$
- In thin plates as a first approximation it may be assumed that:

$$\sigma_{zz} = \sigma_{zx} = \sigma_{zy} = 0$$

- Based on coordinate transformations in 2D:

$$l_1 = \cos\theta; \quad m_1 = \sin\theta; \quad l_2 = -\sin\theta; \quad m_2 = \cos\theta$$

Mohr's Circle in 2D

- Based on stress transformations in 2D:

$$\begin{aligned}\sigma_{x'x'} &= \sigma_{xx}\cos^2\theta + \sigma_{yy}\sin^2\theta + 2\sigma_{xy}\sin\theta\cos\theta\\ \sigma_{y'y'} &= \sigma_{xx}\sin^2\theta + \sigma_{yy}\cos^2\theta - 2\sigma_{xy}sin\theta\cos\theta\\ \sigma_{x'y'} &= -(\sigma_{xx} - \sigma_{yy})\sin\theta cos\theta + \sigma_{xy}(\cos^2\theta - \sin^2\theta)\end{aligned}$$

- Recasting the equations using double angle:

$$\begin{aligned}\sigma_{x'x'} &= \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) + \frac{1}{2}(\sigma_{xx} - \sigma_{yy})\cos 2\theta + \sigma_{xy}\sin 2\theta\\ \sigma_{y'y'} &= \frac{1}{2}(\sigma_{xx} + \sigma_{yy}) - \frac{1}{2}(\sigma_{xx} - \sigma_{yy})\cos 2\theta - \sigma_{xy}\sin 2\theta\\ \sigma_{x'y'} &= -\frac{1}{2}(\sigma_{xx} - \sigma_{yy})\sin 2\theta + \sigma_{xy}\cos 2\theta\end{aligned}$$

- From above set of equations:

$$\sigma_{x'x'} + \sigma_{y'y'} = \sigma_{xx} + \sigma_{yy}; \quad \sigma_{x'x'}\sigma_{y'y'} - \sigma_{x'y'}^2 = \sigma_{xx}\sigma_{yy} - \sigma_{xy}^2$$

Mohr's Circle in 2D

- From above equation set:

$$\left[\sigma_{x'x'} - \frac{1}{2}(\sigma_{xx} + \sigma_{yy})\right]^2 + \sigma_{x'y'}^2 = \frac{1}{4}(\sigma_{xx} + \sigma_{yy})^2 + \sigma_{xy}^2$$

$$\text{Center (C)} \quad : \quad \left(\frac{1}{2}(\sigma_{xx} + \sigma_{yy}), 0\right); \qquad \text{Radius (R)} \quad : \quad \sqrt{\frac{1}{4}(\sigma_{xx} + \sigma_{yy})^2 + \sigma_{xy}^2}$$

- The angle to the principal plane can be defined as:

$$\tan 2\theta = \frac{2\sigma_{xy}}{\sigma_{xx} - \sigma_{yy}}$$

- In terms of Principal Stresses:

$$\sigma_1 = C + R; \quad \sigma_2 = C - R; \quad \tau_{max} = \max(\sigma_{x'y'}) = R$$

Mohr's Circle for 3D Stress States

- Consider principal stresses σ_1, σ_2 and σ_3 where $\sigma_1 \geqslant \sigma_2 \geqslant \sigma_3$.
- Largest shear stress:

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$$

- Consider any plane whose normal has direction cosines relative to the principal axes. On this plane the normal and shear stress components be σ_{NN} & σ_{NS}.
- Using stress transformation rules:

$$\sigma_{NN} = l^2\sigma_1 + m^2\sigma_2 + n^2\sigma_3$$
$$\sigma_{NS}^2 = l^2\sigma_1^2 + m^2\sigma_2^2 + n^2\sigma_3^2 - \sigma_{NN}^2$$
$$l^2 + m^2 + n^2 = 1$$

Mohr's Circle for 3D Stress States

- Solving for direction cosines we obtain:

$$l^2 = \frac{\sigma_{NS}^2 + (\sigma_{NN} - \sigma_2)(\sigma_{NN} - \sigma_3)}{(\sigma_1 - \sigma_2)(\sigma_1 - \sigma_3)} \geqslant 0$$
$$m^2 = \frac{\sigma_{NS}^2 + (\sigma_{NN} - \sigma_1)(\sigma_{NN} - \sigma_3)}{(\sigma_2 - \sigma_1)(\sigma_2 - \sigma_3)} \geqslant 0$$
$$n^2 = \frac{\sigma_{NS}^2 + (\sigma_{NN} - \sigma_1)(\sigma_{NN} - \sigma_2)}{(\sigma_3 - \sigma_1)(\sigma_3 - \sigma_2)} \geqslant 0$$

- From above inequalities we can write:

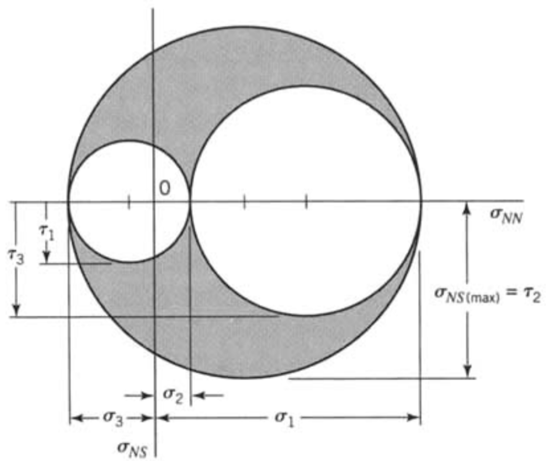
$$\sigma_{NS}^2 + (\sigma_{NN} - \sigma_2)(\sigma_{NN} - \sigma_3) \geqslant 0$$
$$\sigma_{NS}^2 + (\sigma_{NN} - \sigma_3)(\sigma_{NN} - \sigma_1) \leqslant 0$$
$$\sigma_{NS}^2 + (\sigma_{NN} - \sigma_1)(\sigma_{NN} - \sigma_2) \geqslant 0$$

Mohr's Circle for 3D Stress States

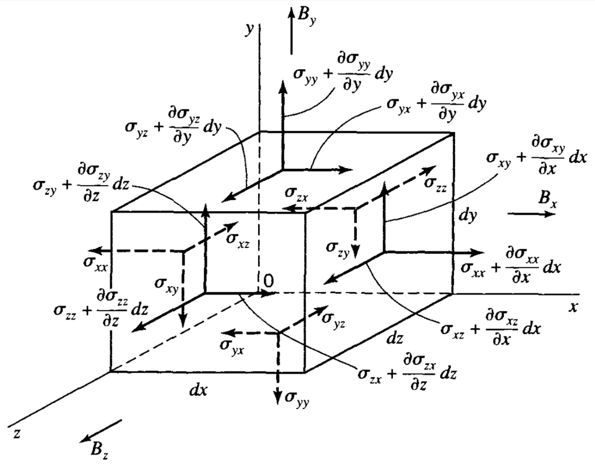
- Above inequalities may be re-written as:

$$\sigma_{NS}^2 + \left(\sigma_{NN} - \frac{\sigma_2 + \sigma_3}{2}\right)^2 \geqslant \frac{1}{4}(\sigma_2 - \sigma_3)^2$$
$$\sigma_{NS}^2 + \left(\sigma_{NN} - \frac{\sigma_3 + \sigma_1}{2}\right)^2 \leqslant \frac{1}{4}(\sigma_3 - \sigma_1)^2$$
$$\sigma_{NS}^2 + \left(\sigma_{NN} - \frac{\sigma_1 + \sigma_2}{2}\right)^2 \geqslant \frac{1}{4}(\sigma_1 - \sigma_2)^2$$

Mohr's Circle for 3D Stress States



Equilibrium Equations



Equilibrium Equations

- Setting up force equilibrium:

$$\sum F_x = \left(\sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x} dx\right) dy\,dz + \left(\sigma_{yx} + \frac{\partial \sigma_{yx}}{\partial y} dy\right) dx\,dz + \left(\sigma_{zx} + \frac{\partial \sigma_{zx}}{\partial z} dz\right) dx\,dy$$

$$- \sigma_{xx}\,dy\,dz - \sigma_{yx}\,dx\,dz - \sigma_{zx}\,dx\,dy + B_x\,dx\,dy\,dz = 0$$

- Differential forms of equilibrium equations:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} + B_x = 0$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + B_y = 0$$

$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + B_z = 0$$

Reading Assignment

Chapter 2: Section 2.1 - 2.3.1